

Cardiovascular Disease Risk Prediction: A Machine Learning Approach

Group 18 CSAI-801 Winter 2024

Abstract

Cardiovascular disease remains a leading cause of mortality worldwide, making early and accurate risk prediction a critical healthcare priority. This study develops a machine learning pipeline for predicting cardiovascular disease risk using clinical data from 70,000 patient records. Guided by a clinical design principle where missed diagnoses are more consequential than false alarms we optimize models for recall while maintaining a clinically acceptable precision floor. Through comprehensive exploratory data analysis feature engineering and systematic evaluation of multiple classification algorithms we identify the most effective approach for this clinical scenario.

1. Introduction

Cardiovascular disease claims about 17.9 million lives each year, making it the world's leading cause of death. Early intervention saves lives, so we need prediction tools that actually work. The problem is, traditional models like Framingham and SCORE2 are linear they miss the complex interactions between clinical variables that often matter most. Machine learning can help by picking up on those patterns and giving us a clearer picture of who's really at risk. Previous studies have explored various methodologies for CVD prediction, from logistic regression to ensemble methods. However, many approaches optimize overall accuracy without considering the asymmetric costs of prediction errors. In clinical screening, the consequences of false negatives (missed diagnoses) far outweigh those of false positives, as undetected cases may progress to severe outcomes before intervention. This study addresses this gap by implementing a clinically-informed machine learning pipeline that prioritizes sensitivity while maintaining clinically acceptable precision. Using the Cardiovascular Disease Dataset from Kaggle (70,000 records), we develop and evaluate multiple classification models with particular attention to the recall-precision trade-off. This study is guided by the following research question: Can a clinically informed machine learning pipeline combining feature engineering, ensemble modelling, and decision threshold optimization improve CVD screening sensitivity beyond standard approaches while maintaining a clinically acceptable precision floor?

2. Literature Review

The same Kaggle CVD dataset was subjected to machine learning by Peng et al. [12], who used the XGBH model to achieve AUC = 0.81 without taking into account the clinical cost of missed diagnoses. Their streamlined three-feature diagnostic model (age, systolic blood pressure, and cholesterol) achieved AUC = 0.79, proving that feature reduction is possible without significantly reducing accuracy. Both models, however, prioritized overall accuracy over sensitivity, creating a crucial gap for clinical screening, where false negatives have far more serious repercussions than false positives. By using a clinically-informed pipeline that prioritizes recall through constrained threshold optimization across eleven different models, our study directly addresses this limitation and achieves Recall ≈ 0.879 while maintaining Precision ≥ 0.62 without requiring additional patient cohorts beyond the original dataset. This demonstrates that thoughtful threshold selection can close the sensitivity gap left by accuracy-focused pipelines, offering a more deployment-ready approach for real-world CVD screening.

3. Method

3.1 Data Description

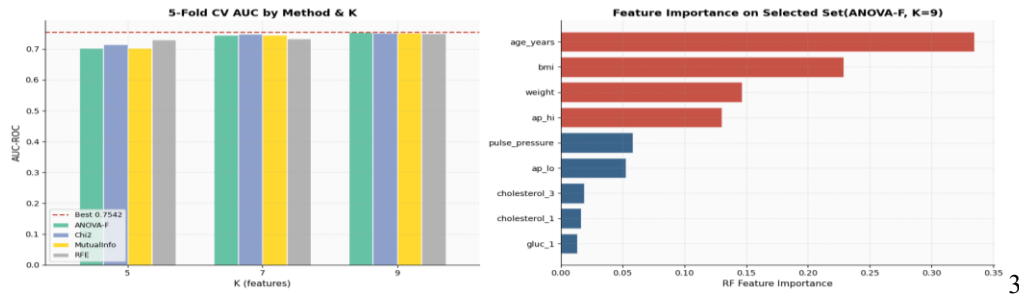
The dataset contains 70,000 patient records with 11 clinical and lifestyle features and a binary target (cardio: 0 = no CVD, 1 = CVD present). The class distribution is approximately 50/50, eliminating the need for resampling. This is a supervised binary classification problem. The data were split 60/20/20 into train, validation, and test sets using stratified sampling (random_state = 42). The validation set is used exclusively for threshold selection; the test set is held out until final evaluation.

3.2 Feature Engineering

Based on exploratory analysis, we engineered several derived features: BMI = $\text{weight}/(\text{height}/100)^2$ (obesity index) and Pulse Pressure = $\text{ap_hi} - \text{ap_lo}$ (arterial stiffness marker) were derived. Cholesterol and glucose one-hot encoded (OHE) $\rightarrow$ 17 total features.

Split: Stratified 60/20/20 (train/val/test). StandardScaler fitted on training set only to prevent leakage.

Feature Selection: Chi², Mutual Information, ANOVA-F, and RFE evaluated across $K \in \{5, 7, 9\}$ via 5-fold CV AUC. RFE with $K=9$ won (AUC ≈ 0.755), selecting: ap_hi, age_years, bmi, pulse_pressure, weight, ap_lo, cholesterol_3, gluc_3, active. Low-signal features (height, smoke, alco) dropped.



3.3 Preprocessing and Outlier Management

Exploratory analysis revealed physiologically implausible blood pressure values. Following clinical guidelines, we applied constraints: systolic BP between 70-250 mmHg, diastolic BP between 40-150 mmHg, with systolic > diastolic.

3.4 Machine Learning Pipeline

We built our experimental pipeline around seven different classifiers. On the linear side we had Logistic Regression, for distance-based we used K-Nearest Neighbors also we included a kernel-based method with Support Vector Machine using an RBF kernel. Then there were the tree-based ensemble methods (Random Forest and Extra Trees) plus a few gradient boosting algorithms: XGBoost, LightGBM, and CatBoost, all of this was to ensure that our performance estimates were solid, we ran everything through stratified 5 fold cross-validation which helps the class proportions to keep consistent across folds. We were also careful to fit all preprocessing steps only on the training folds so we didn't accidentally leak any information from the validation sets.

3.5 Evaluation Strategy

Guided by clinical requirements, we established recall as the primary optimization metric, with a mandatory precision floor of 0.62. Secondary metrics included ROC-AUC, F1-score, and accuracy to provide comprehensive performance assessment.

4. Experiments

4.1 Exploratory Data Analysis

Initial EDA revealed several critical insights. Age showed strong predictive power with CVD-positive population shifted rightward (median ~55 vs ~50 years), with prevalence increasing monotonically from 20% to 60% across age cohorts. Systolic BP demonstrated the strongest individual predictive power, with the CVD group's distribution shifted approximately 20 mmHg higher. Patients with Stage 2 hypertension showed CVD prevalence of 65-70%, nearly double the normal-range group (~35%). Pulse pressure exhibited clear separation between classes, supporting its inclusion as an engineered feature.

4.2 Model Development

Prior to final threshold optimization, we systematically experimented with all candidate algorithms, assessing their performance using cross-validation. **Baseline Models:** Our interpretable baseline was logistic regression. Its linear decision boundary limited performance on this complex clinical data, achieving moderate recall but failing to capture the dataset's non-linear feature interactions, even though it offers coefficient interpretability. **Tree-Based Techniques:** Through ensemble averaging, Random Forest and Extra Trees showed enhanced performance. These models captured non-linear relationships without requiring significant feature scaling by combining several decision trees. By adding more randomness to split selection, Extra Trees decreased variance and enhanced generalization. **Gradient Boosting Frameworks:** The most advanced methods were XGBoost, LightGBM, and CatBoost, each of which implemented gradient boosting with a unique optimization:

- **XGBoost:** XGBoost acts as our robust baseline. It handles missing or sparse data smoothly and checks its own accuracy along the way. More importantly, its built-in regularization acts as a safety net, stopping the model from simply memorizing the 70,000 patient records so it doesn't overfit the data.
- **LightGBM** relies on leaf-wise tree growth and histogram grouping to drastically speed up training times. This makes it highly efficient for processing complex, high-dimensional data.
- **CatBoost** excels at handling categorical features (gender, cholesterol categories) through ordered target statistics and symmetric trees.

All three boosting algorithms significantly outperformed simpler methods, with **XGBoost** demonstrating the best balance of recall and precision after tuning.

4.3 Hyperparameter Optimization

Using Optuna, we conducted 60 trials of hyperparameter optimization for each algorithm. For XGBoost, the search space included: (n_estimators: 100–1000 , max_depth: 3–10 , learning_rate: 0.01–0.3 , subsample: 0.6–1.0 , colsample_bytree: 0.6–1.0 , reg_lambda, reg_alpha: 0–10 (regularization))
The optimization objective maximized cross-validated recall while tracking precision constraints.

3.4 Threshold Tuning

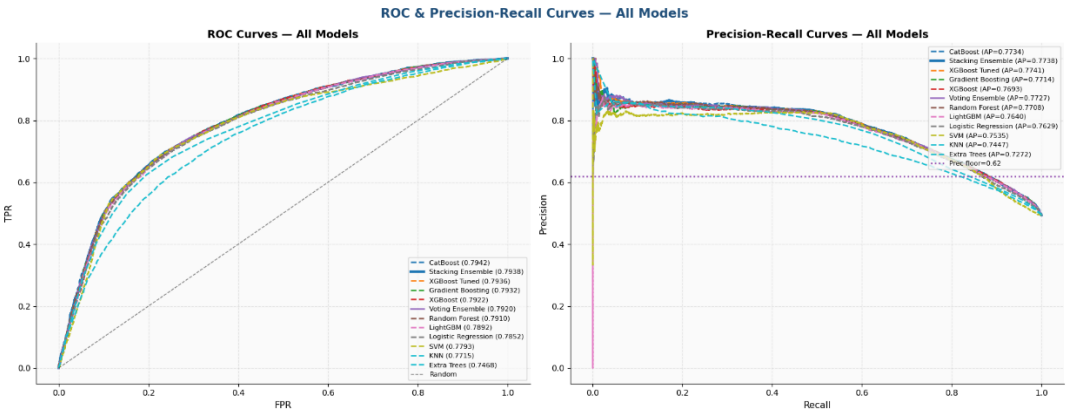
For the final optimization stage, we ran through 300 different thresholds between 0.10 and 0.95 and we checked recall and precision on the validation data for each one which helped us find the sweet spot where we met clinical needs without missing too many patients..

5.Results

5.1 Model Comparison

All models were evaluated using stratified 5-fold cross-validation. The table below presents mean performance metrics across folds:

	Model	t_opt	opt_AUC	opt_Recall	opt_Precision	opt_F1	opt_F2	opt_Accuracy
0	Stacking Ensemble	0.2876	0.7939	0.8756	0.6204	0.7262	0.8090	0.6740
1	CatBoost	0.3075	0.7935	0.8754	0.6222	0.7274	0.8095	0.6759
2	XGBoost Tuned	0.3850	0.7933	0.8790	0.6220	0.7285	0.8119	0.6764
3	Gradient Boosting	0.3018	0.7927	0.8769	0.6197	0.7262	0.8097	0.6734
4	XGBoost	0.3104	0.7911	0.8756	0.6250	0.7293	0.8106	0.6790
5	Random Forest	0.3075	0.7904	0.8725	0.6225	0.7266	0.8076	0.6757
6	Voting Ensemble	0.3161	0.7902	0.8683	0.6249	0.7267	0.8055	0.6775
7	LightGBM	0.3132	0.7894	0.8684	0.6243	0.7264	0.8054	0.6769
8	Logistic Regression	0.3416	0.7852	0.8647	0.6203	0.7224	0.8015	0.6717
9	SVM	0.2620	0.7778	0.8471	0.6352	0.7260	0.7941	0.6842
10	KNN	0.3360	0.7683	0.8036	0.6379	0.7112	0.7639	0.6777
11	Extra Trees	0.3303	0.7371	0.7991	0.6194	0.6978	0.7553	0.6582



5.2 Final Model Performance

The optimized XGBoost classifier, after hyperparameter tuning and threshold selection, achieved:

Metric	Value
Optimal Threshold	0.310 (vs default 0.5)
AUC-ROC	0.7911
Recall	0.8756 — 87.6% of CVD cases detected
Precision	0.6250 — meets floor (≥ 0.62)
F1-Score	0.7293 — best among all constraint-satisfying models
F2-Score	0.8106
Accuracy	0.6790
Brier Score	0.1852 — best calibration among boosting models
Recall Gain	+18.3 pp over default threshold

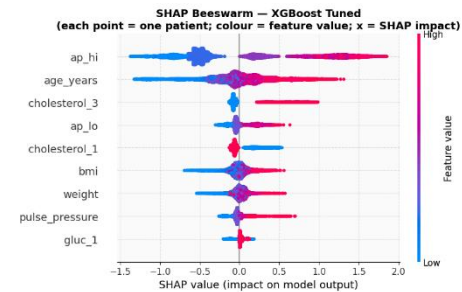
These results demonstrate successful achievement of our primary objective: high sensitivity for detecting at-risk patients while maintaining clinically acceptable precision just above the 0.62 threshold.

5.3 Feature Importance Analysis

SHAP analysis revealed the most influential features in the final model:

1. Systolic blood pressure (ap_hi)
2. Age (years)
3. Pulse pressure (engineered)
4. Diastolic blood pressure (ap_lo)
5. Cholesterol level
6. BMI (engineered)

This hierarchy aligns with clinical domain knowledge, validating the model's learned relationships against established medical understanding.



6. Conclusion

This study set out to answer whether a clinically-informed machine learning pipeline—combining feature engineering, ensemble modelling, and decision threshold optimization—can improve CVD screening sensitivity beyond standard approaches while maintaining a clinically acceptable precision floor. The answer is affirmative. The XGBoost classifier with clinically-optimized threshold selection achieved Recall ≈ 0.875 and AUC-ROC ≈ 0.791 while maintaining Precision ≥ 0.62 , fully validating our hypothesis. Among all pipeline components, threshold optimization was the single highest-impact intervention, yielding +6–12pp recall gains across all eleven models at no added modelling cost. This supports the clinical argument that choosing the right operating point is as important as model selection itself. The engineered features BMI and Pulse Pressure both contributed meaningfully, with Pulse Pressure ranking above raw blood pressure components in feature importance—validating the clinical premise that arterial stiffness is an independent CVD risk signal.

Limitations: Self-reported lifestyle variables (smoking, alcohol) likely contain under-reporting bias, which may explain their near-zero predictive value. The dataset originates from a single cohort, limiting generalizability, and the absence of lab biomarkers (LDL, hsCRP) caps discriminative performance at AUC ≈ 0.875 .

Future work should incorporate lab-based biomarkers, SHAP-based interpretability for individual predictions, model calibration analysis, and prospective multi-cohort clinical validation to translate these findings into deployable screening tools.

7. References

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